



# API Guide User Manual.

**Version 1.0.0**

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# 1 Introduction

## 1.1 What is TV White Space?

TV White Space (TVWS) refers to unused portions of radio-frequency spectrum in the television broadcasting bands (VHF/UHF, roughly 54–698 MHz). These spectrum gaps arise geographically and temporally from the digital television transition and the uneven distribution of broadcast transmitters across a territory.

TVWS technology enables *secondary* devices—such as broadband internet access equipment—to operate on idle TV channels without interfering with *primary* users (licensed television broadcasters and wireless microphones). Secondary access is coordinated through a geolocation database, which tracks which TV channels are available at any given location and time.

## 1.2 The TVWS-DB API

The TVWS-DB API is a server-side implementation of the Protocol to Access White-Space (PAWS) Databases, as specified in **RFC 7545** Mancuso et al., 2015. It exposes a single HTTP endpoint through which TVWS devices query spectrum availability for a given geographic location.

The API performs the following functions:

- Validates device identity (manufacturer, model, serial number, ruleset).
- Verifies the geographic coordinates of the requesting device.
- Enforces antenna parameter constraints (height, direction, radiation pattern).
- Returns the list of available TV channels and their maximum permitted transmission power for the queried location.

## 1.3 RFC 7545 Compliance

This API fully implements the PAWS protocol defined in RFC 7545 Mancuso et al., 2015. The PAWS framework specifies:

- A **JSON-RPC 2.0** message transport model over HTTP.
- Three mandatory request/response message types: `INIT_REQ`, `REGISTRATION_REQ`, and `AVAIL_SPECTRUM_REQ`.
- A standardized error reporting model with well-defined numeric error codes.
- Ruleset-based spectrum availability rules (e.g., `FccTvBandWhiteSpace-2010`).

In addition to the standard PAWS error codes, this implementation defines a set of extension codes for authentication and session management (see Section 4).

## 1.4 Base Endpoint

All PAWS requests are sent as HTTP **POST** requests to:

POST /paws

The request body must be a valid JSON-RPC 2.0 object. The Content-Type header must be application/json.

### Health Check

A lightweight health check endpoint is also available:

GET /paws/test

Returns HTTP 200 with body "up" when the server is running.

## 2 Usage Flow

### 2.1 Overview

The PAWS protocol defines a mandatory three-step sequence. Every TVWS device must complete these steps **in order** before obtaining spectrum availability data:

1. **INIT\_REQ** — Device initialization. The database acknowledges the device and returns ruleset information for the device's location.
2. **REGISTRATION\_REQ** — Device registration. The device provides owner and operator contact information (vCard format).
3. **AVAIL\_SPECTRUM\_REQ** — Spectrum availability query. The device sends its precise location and antenna parameters; the database returns the available TV channels with permitted power levels.

### 2.2 JSON-RPC 2.0 Message Structure

All PAWS messages use the JSON-RPC 2.0 envelope. Every request must include the four top-level fields shown below.

| Field   | Type   | Description                                           |
|---------|--------|-------------------------------------------------------|
| jsonrpc | string | Protocol version. Must be "2.0".                      |
| method  | string | PAWS method name (varies per message type).           |
| params  | object | Message-specific parameters.                          |
| id      | string | Client-assigned request identifier (e.g., "2026.1a"). |

A successful response contains a `result` field. An error response contains an `error` field with the sub-fields `code`, `message`, and `data.parameters`.

## 2.3 Step 1 — INIT\_REQ

**Method:** `spectrum.paws.init`

The initialization request introduces the device to the database. The device provides its identity (`deviceDesc`) and current location. On success, the database returns an `INIT_RESP` containing the applicable ruleset information.

## 2.4 Step 2 — REGISTRATION\_REQ

**Method:** `spectrum.paws.register`

The registration request extends the initialization message with owner and operator contact information in jCard (vCard 4.0) format. This step must be completed once before any spectrum query is accepted.

## 2.5 Step 3 — AVAIL\_SPECTRUM\_REQ

**Method:** `spectrum.paws.getSpectrum`

This is the primary query. The device provides its current location, antenna parameters, and optionally master device information. The database returns the list of available TV channels with their maximum permitted transmission power for the queried location.

This request can be repeated as often as needed (subject to polling limits defined in the `INIT_RESP` ruleset). The device must re-send registration data (`deviceOwner`) with each request.

# 3 Message Reference

This section documents the complete field reference for each PAWS message type. Fields are expressed in dot notation (e.g., `params.location.point.center.latitude`).

The **Req.** column uses:

- **Yes** — field must be present and valid.
- **No** — field is optional.
- **Cond.** — required only when a related field is present (see Constraints column).

## 3.1 INIT\_REQ

**Method:** `spectrum.paws.init`

Listing 1: `INIT_REQ` — complete example

```
1 {  
2   "jsonrpc": "2.0",  
3   "method": "spectrum.paws.init",  
4   "params": {  
5     "type": "INIT_REQ",  
6     "version": "1.0",
```

```

7  "deviceDesc": {
8    "serialNumber": "4fbf-bae3-c8c5706745e3",
9    "fccId": "94d8610a-9eea",
10   "manufacturerId": "123-ABCD-4fbf-bae3-c8c5706745e3",
11   "modelId": "xpto-123-123-12",
12   "rulesetIds": ["FccTvBandWhiteSpace-2010"]
13 },
14 "location": {
15   "point": {
16     "center": {
17       "latitude": -3.752542,
18       "longitude": -38.556012
19     }
20   }
21 },
22 },
23 "id": "2026.1a"
24 }
```

| Field                                  | Type     | Req. | Constraints                          |
|----------------------------------------|----------|------|--------------------------------------|
| jsonrpc                                | string   | Yes  | Must be "2.0"                        |
| method                                 | string   | Yes  | "spectrum.paws.init"                 |
| id                                     | string   | Yes  | Request identifier                   |
| params.type                            | string   | Yes  | "INIT_REQ"                           |
| params.version                         | string   | Yes  | "1.0"                                |
| params.deviceDesc.serialNumber         | string   | Yes  | Device serial number                 |
| params.deviceDesc.fccId                | string   | Yes  | FCC identifier                       |
| params.deviceDesc.manufacturerId       | string   | Yes  | Manufacturer identifier              |
| params.deviceDesc.modelId              | string   | Yes  | Model identifier                     |
| params.deviceDesc.rulesetIds           | string[] | Yes  | e.g.<br>["FccTvBandWhiteSpace-2010"] |
| params.location.point.center.latitude  | number   | Yes  | Range: [-90, 90]                     |
| params.location.point.center.longitude | number   | Yes  | Range: [-180, 180]                   |

## 3.2 REGISTRATION\_REQ

**Method:** spectrum.paws.register

REGISTRATION\_REQ includes all fields from INIT\_REQ (Section 3.1), with method and params.type changed as shown, plus the additional deviceOwner object.

Listing 2: REGISTRATION\_REQ — example (deviceOwner section)

```

1 {
2   "jsonrpc": "2.0",
```

```
3  "method": "spectrum.paws.register",
4  "params": {
5      "type": "REGISTRATION_REQ",
6      "version": "1.0",
7      "deviceDesc": {
8          "serialNumber": "4fbf-bae3-c8c5706745e3",
9          "fccId": "94d8610a-9eea",
10         "manufacturerId": "123-ABCD-4fbf-bae3-c8c5706745e3",
11         "modelId": "xpto-123-123-123",
12         "rulesetIds": [
13             "FccTvBandWhiteSpace-2010"
14         ]
15     },
16     "location": {
17         "point": {
18             "center": {
19                 "latitude": -4.978806,
20                 "longitude": -39.056607
21             }
22         }
23     },
24     "deviceOwner": {
25         "owner": {
26             "vcard": [
27                 [
28                     "version",
29                     {
30
31                     },
32                     "text",
33                     "4.0"
34                 ],
35                 [
36                     "fn",
37                     {
38
39                     },
40                     "text",
41                     "Fractal Networks"
42                 ],
43                 [
44                     "org",
45                     {
46
47                     },
48                     "text",
49                     "Fractal Networks"
50                 ]
51             ]
52         }
53     }
54 }
```

```
51     [
52         "adr",
53         {
54
55         },
56         "text",
57         [
58             "",
59             "",
60             "123 Innovation Ave",
61             "San Francisco",
62             "CA",
63             "94107",
64             "USA"
65         ]
66     ],
67     [
68         "email",
69         {
70
71         },
72         "text",
73         "contact@fractalfn.com"
74     ],
75     [
76         "tel",
77         {
78
79         },
80         "uri",
81         "tel:+1-415-555-0100"
82     ]
83 ],
84 },
85 "operator":{
86     "vcard":[
87         [
88             "version",
89             {
90
91             },
92             "text",
93             "4.0"
94         ],
95         [
96             "fn",
97             {
```



```
99      },
100      "text",
101      "Jane Doe"
102    ],
103    [
104      "org",
105      {
106
107      },
108      "text",
109      "Fractal Networks"
110    ],
111    [
112      "adr",
113      {
114
115      },
116      "text",
117      [
118        "",
119        "",
120        "123 Innovation Ave",
121        "San Francisco",
122        "CA",
123        "94107",
124        "USA"
125      ]
126    ],
127    [
128      "email",
129      {
130
131      },
132      "text",
133      "jane.doe@fractalfn.com"
134    ],
135    [
136      "tel",
137      {
138
139      },
140      "uri",
141      "tel:+1-415-555-0101"
142    ]
143  ]
144 }
145 }
146 },
```

```
147 "id": "10"
148 }
```

| Field                                                | Type   | Req. | Constraints                      |
|------------------------------------------------------|--------|------|----------------------------------|
| method                                               | string | Yes  | "spectrum.paws.register"         |
| params.type                                          | string | Yes  | "REGISTRATION_REQ"               |
| <i>All INIT_REQ fields also apply (Section 3.1).</i> |        |      |                                  |
| params.deviceOwner.owner.vcard                       | array  | Yes  | jCard 4.0 array (see note below) |
| params.deviceOwner.operator.vcard                    | array  | Yes  | jCard 4.0 array (see note below) |

**vCard format.** The vcard field follows the jCard specification (RFC 7095). Each entry is a four-element array: [property, parameters, type, value]. Supported properties: version (must be "4.0"), fn (full name), org (organization), adr (address), email, and tel.

### 3.3 AVAIL\_SPECTRUM\_REQ

**Method:** spectrum.paws.getSpectrum

AVAIL\_SPECTRUM\_REQ includes all fields from REGISTRATION\_REQ (Section 3.2), with method and params.type changed as shown, plus the mandatory antenna object and optional master device fields.

Listing 3: AVAIL\_SPECTRUM\_REQ — antenna and master device fields

```
1 {
2   "jsonrpc": "2.0",
3   "method": "spectrum.paws.getSpectrum",
4   "params": {
5     "type": "AVAIL_SPECTRUM_REQ",
6     "version": "1.0",
7     "deviceDesc": {
8       "serialNumber": "4fbf-bae3-c8c5706745e3",
9       "fccId": "94d8610a-9eea",
10      "manufacturerId": "123-ABCD-4fbf-bae3-c8c5706745e3",
11      "modelId": "xpto-123-123-123",
12      "rulesetIds": [
13        "FccTvBandWhiteSpace-2010"
14      ]
15    },
16    "masterDeviceDesc": {
17      "serialNumber": "4fbf-bae3-c8c5706745e3",
18      "fccId": "94d8610a-9eea",
19      "manufacturerId": "123-ABCD-4fbf-bae3-c8c5706745e3",
```

```
20     "modelId": "xpto-123-123-123",
21     "rulesetIds": [
22         "FccTvBandWhiteSpace-2010"
23     ]
24 },
25 "location": {
26     "point": {
27         "center": {
28             "latitude": -3.7653855,
29             "longitude": -38.5261224
30         }
31     }
32 },
33 "deviceOwner": {
34     "owner": {
35         "vcard": [
36             [
37                 "version",
38                 {
39
40                 },
41                 "text",
42                 "4.0"
43             ],
44             [
45                 "fn",
46                 {
47
48                 },
49                 "text",
50                 "Fractal Networks"
51             ],
52             [
53                 "org",
54                 {
55
56                 },
57                 "text",
58                 "Fractal Networks"
59             ],
60             [
61                 "adr",
62                 {
63
64                 },
65                 "text",
66                 [
67                     "",
```

```
68         "",
69         "123 Innovation Ave",
70         "San Francisco",
71         "CA",
72         "94107",
73         "USA"
74     ],
75 ],
76 [
77     "email",
78     {
79
80     },
81     "text",
82     "contact@fractalfn.com"
83 ],
84 [
85     "tel",
86     {
87
88     },
89     "uri",
90     "tel:+1-415-555-0100"
91 ]
92 ],
93 },
94 "operator":{
95     "vcard":[
96         [
97             "version",
98             {
99
100             },
101             "text",
102             "4.0"
103         ],
104         [
105             "fn",
106             {
107
108             },
109             "text",
110             "Jane Doe"
111         ],
112         [
113             "org",
114             {
```

```
116         },
117         "text",
118         "Fractal Networks"
119     ],
120     [
121         "adr",
122         {
123
124         },
125         "text",
126         [
127             "",
128             "",
129             "123 Innovation Ave",
130             "San Francisco",
131             "CA",
132             "94107",
133             "USA"
134         ]
135     ],
136     [
137         "email",
138         {
139
140         },
141         "text",
142         "jane.doe@fractalfn.com"
143     ],
144     [
145         "tel",
146         {
147
148         },
149         "uri",
150         "tel:+1-415-555-0101"
151     ]
152 ]
153 }
154 },
155 "antenna":{
156     "height":70,
157     "heightType":"AGL",
158     "antennaDirection":{
159         "zenith":10.1,
160         "azimuth":12.1
161     },
162     "antennaRadiationPattern":[
163
```

|     |      |
|-----|------|
| 164 | 0.0, |
| 165 | 0.0, |
| 166 | 0.0, |
| 167 | 0.0, |
| 168 | 0.0, |
| 169 | 0.0, |
| 170 | 0.0, |
| 171 | 0.0, |
| 172 | 0.0, |
| 173 | 0.0, |
| 174 | 0.0, |
| 175 | 0.0, |
| 176 | 0.0, |
| 177 | 0.0, |
| 178 | 0.0, |
| 179 | 0.0, |
| 180 | 0.0, |
| 181 | 0.0, |
| 182 | 0.0, |
| 183 | 0.0, |
| 184 | 0.0, |
| 185 | 0.0, |
| 186 | 0.0, |
| 187 | 0.0, |
| 188 | 0.0, |
| 189 | 0.0, |
| 190 | 0.0, |
| 191 | 0.0, |
| 192 | 0.0, |
| 193 | 0.0, |
| 194 | 0.0, |
| 195 | 0.0, |
| 196 | 0.0, |
| 197 | 0.0, |
| 198 | 0.0, |
| 199 | 0.0, |
| 200 | 0.0, |
| 201 | 0.0, |
| 202 | 0.0, |
| 203 | 0.0, |
| 204 | 0.0, |
| 205 | 0.0, |
| 206 | 0.0, |
| 207 | 0.0, |
| 208 | 0.0, |
| 209 | 0.0, |
| 210 | 0.0, |
| 211 | 0.0, |

```

212         0.0,
213         0.0,
214         0.0,
215         0.0,
216         0.0,
217         0.0,
218         0.0,
219         0.0,
220         0.0,
221         0.0,
222         0.0,
223         0.0,
224         0.0,
225         0.0,
226         0.0,
227         0.0,
228         0.0,
229         0.0,
230         0.0,
231         0.0,
232         0.0,
233         0.0,
234         0.0,
235         0.0
236     ],
237 ],
238     "antennaPolarization":[
239         "h"
240     ],
241 },
242     "masterDeviceLocation":{
243         "point":{
244             "center":{
245                 "latitude":-3.7653855,
246                 "longitude":-38.5261224
247             }
248         }
249     },
250 },
251     "id":"10"
252 }

```

| Field       | Type   | Req. | Constraints                 |
|-------------|--------|------|-----------------------------|
| method      | string | Yes  | "spectrum.paws.getSpectrum" |
| params.type | string | Yes  | "AVAIL_SPECTRUM_REQ"        |

(continued)

| Field                                                        | Type       | Req.  | Constraints                                                                                                   |
|--------------------------------------------------------------|------------|-------|---------------------------------------------------------------------------------------------------------------|
| <i>All REGISTRATION_REQ fields also apply (Section 3.2).</i> |            |       |                                                                                                               |
| params.antenna.height                                        | number     | Yes   | AGL: [−200, 2000] m<br>AMSL: [−3500, 8848] m                                                                  |
| antenna.heightType                                           | string     | Yes   | "AGL" or "AMSL"                                                                                               |
| antenna.antennaDirection.zenith                              | number     | Cond. | [−90, 90]; required when antennaRadiationPattern is present                                                   |
| antenna.antennaDirection.azimuth                             | number     | Cond. | [0, 360]; required when antennaRadiationPattern is present                                                    |
| antenna.antennaPolarization                                  | string[]   | Cond. | ["h"], ["v"], or ["h", "v"]; required when antennaRadiationPattern is present                                 |
| antenna.antennaRadiationPattern                              | number[][] | No    | 1 sub-array (single pol.) or 2 sub-arrays (dual pol.); each must contain exactly <b>72</b> numeric values     |
| params.antenna.antennaGain                                   | number     | No    | Antenna gain in dBi; valid range: [0, 50] dBi. Returns error −202 if present but out of range or not a number |
| params.masterDeviceDesc                                      | object     | No    | Same structure as params.deviceDesc                                                                           |
| params.masterDeviceLocation                                  | object     | No    | Same structure as params.location                                                                             |

### Height types.

- **AGL** (Above Ground Level): height measured from the ground surface directly below the antenna. Valid range: −200 to 2000 metres.
- **AMSL** (Above Mean Sea Level): height measured from mean sea level. Valid range: −3500 to 8848 metres (Dead Sea floor to Mt. Everest peak).

**Antenna radiation pattern.** When antennaRadiationPattern is present, both antennaDirection and antennaPolarization become mandatory. The number of sub-arrays must match the polarization count: **1** sub-array for single polarization (["h"] or ["v"]), **2** sub-arrays for dual polarization (["h", "v"]). Each sub-array must contain exactly 72 numeric values representing the antenna gain (in dBi) at 5° intervals from 0° to 355°. Mismatches between polarization count and sub-array count return error −201.



**Antenna gain (antennaGain).** The optional `antennaGain` field inside the `antenna` object allows the device to report the peak gain of its antenna. When present, the compliance module (`server/messages/compliance.js`) validates this field before any spectrum query is processed:

- The value must be a finite number (not a string, boolean, or `null`).
- The value must fall within the range `[0, 50]` dBi, as configured by `antennaParameters.minGainDbi` and `antennaParameters.maxGainDbi` in `config/default.json`.
- If either condition is violated, the server immediately returns error `-202 (INVALID_VALUE)` with the offending field path `params.antenna.antennaGain` and the request is not forwarded to the spectrum service.
- If the field is absent, validation is skipped and the request proceeds normally.

The accepted range (0–50 dBi) covers the full span of practical TV white-space antenna designs, from isotropic reference antennas (0 dBi) to high-gain directional arrays (up to 50 dBi). Values outside this range are considered physically unrealistic and are rejected to prevent misconfigured devices from obtaining incorrect spectrum allocations.

## 4 Error Codes

When a request fails validation or cannot be processed, the API returns a JSON-RPC 2.0 error response. The error object structure is:

Listing 4: Error response structure

```
1 {  
2   "jsonrpc": "2.0",  
3   "error": {  
4     "code": -202,  
5     "message": "A parameter value is invalid in some way.",  
6     "data": {  
7       "parameters": ["params.antenna.height"]  
8     }  
9   },  
10  "id": "2026.1a"  
11 }
```

The `data.parameters` array lists the specific fields that triggered the error. A successful response contains a `result` field instead of `error`.

Error codes are grouped by numeric range. Codes in the `-1xx` and `-2xx` ranges are defined by RFC 7545 Mancuso et al., 2015. Codes in the `-3xx` and `-5xx` ranges are API extensions.

## 4.1 Protocol Errors — *-1xx*

Returned when the request cannot be understood or the device/ruleset is not supported.

| Code | Name                 | Origin    | Description                                                                                                           |
|------|----------------------|-----------|-----------------------------------------------------------------------------------------------------------------------|
| -101 | UNSUPPORTED          | RFC 7545  | The database does not support the specified protocol version ( <code>params.version</code> ).                         |
| -102 | UNSUPPORTED          | RFC 7545  | The database does not support the device type or the specified ruleset ( <code>params.deviceDesc.rulesetIds</code> ). |
| -103 | UNIMPLEMENTED        | RFC 7545  | The database does not implement the requested optional feature.                                                       |
| -104 | OUTSIDE_COVERAGE     | RFC 7545  | The device's location is outside the database's coverage area.                                                        |
| -105 | DATABASE_CHANGE      | RFC 7545  | The database URI has changed. The device must update its list of available databases.                                 |
| -106 | SESSION_ERROR        | Extension | An error occurred in the session layer.                                                                               |
| -107 | DATABASE_UNAVAILABLE | Extension | The database backend is not available at this time.                                                                   |

## 4.2 Parameter Errors — *-2xx*

Returned when request parameters are missing or contain invalid values. These are the most common errors during integration.

| Code | Name          | Origin   | Description                                                                                                                                                                                                |
|------|---------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -201 | MISSING       | RFC 7545 | A required parameter is absent from the request. The <code>data.parameters</code> array lists each missing field path.                                                                                     |
| -202 | INVALID_VALUE | RFC 7545 | A parameter is present but its value is invalid: wrong type, out of permitted range, or logically inconsistent with another field. The <code>data.parameters</code> array identifies the offending fields. |

### Common -202 triggers

| Field                                                | Condition that triggers -202                                          |
|------------------------------------------------------|-----------------------------------------------------------------------|
| <code>params.type</code>                             | Value does not match the method field.                                |
| <code>params.location.point.center.latitude</code>   | Not a number, or outside $[-90, 90]$ .                                |
| <code>params.location.point.center.longitude</code>  | Not a number, or outside $[-180, 180]$ .                              |
| <code>params.antenna.height</code>                   | Not a number, or out of range for the given <code>heightType</code> . |
| <code>params.antenna.heightType</code>               | Not "AGL" or "AMSL".                                                  |
| <code>params.antenna.antennaDirection.azimuth</code> | Not a number, or outside $[0, 360]$ .                                 |
| <code>params.antenna.antennaDirection.zenith</code>  | Not a number, or outside $[-90, 90]$ .                                |
| <code>params.antenna.antennaDirection</code>         | Absent when <code>antennaRadiationPattern</code> is present.          |
| <code>params.antenna.antennaPolarization</code>      | Absent when <code>antennaRadiationPattern</code> is present.          |
| <code>params.antenna.antennaRadiationPattern</code>  | Not an array, or a sub-array element is not a number.                 |

## 4.3 Authentication and Access Errors — -3xx

Returned when authentication or session checks fail. These codes are API extensions to RFC 7545.

| Code | Name                  | Origin    | Description                                                                           |
|------|-----------------------|-----------|---------------------------------------------------------------------------------------|
| -301 | UNAUTHORIZED          | Extension | The request is not authorized.                                                        |
| -302 | NOT_REGISTERED        | RFC 7545  | The device must complete <code>REGISTRATION_REQ</code> before sending other requests. |
| -303 | ACCESS_EXPIRED        | Extension | The device's session or access token has expired. Re-authentication is required.      |
| -304 | AUTHENTICATION_FAILED | Extension | The provided authentication credentials are invalid or missing.                       |

## 4.4 Server Errors — -5xx

| Code | Name         | Origin    | Description                                          |
|------|--------------|-----------|------------------------------------------------------|
| -500 | SERVER_ERROR | Extension | An unexpected internal error occurred on the server. |

## References

Mancuso, V., Probasco, S., & Patil, B. (2015, May). *Protocol to Access White-Space (PAWS) Databases* (RFC No. 7545). Internet Engineering Task Force (IETF). <https://doi.org/10.17487/RFC7545>